MAX KOTAS

Electrical Engineer

McKinney, TX | Willing to relocate to Austin, TX
469-601-3652 | max@maxkotas.com

Electrical engineer designing multilayer mixed-signal PCBAs from customer requirements through production support. Hands-on experience in hardware validation, root cause analysis, and rapid prototyping of embedded electromechanical systems.

EXPERIENCE

Panasonic Energy 2025 - Present
Associate Electrical Engineer | Smart Battery Systems The Colony, TX

- Own battery management system (BMS) schematic design for infusion pumps, AEDs, and solar/data-center energy storage. Translate customer specifications into electrical requirements and select components for protection, power delivery, and communication interfaces.
- Design mixed-signal battery management and power distribution PCBAs, integrating primary/secondary protection ICs, charge/discharge MOSFETs, current sensing and RC filtering, thermistors, self-blow fuses, and filtered SMBus interfaces.
- Support 20+ PCB builds spanning fabrication, prototypes, preproduction, and mass production. Review component derating and test-point coverage, and coordinate with PCB vendors on fabrication and qualification.
- Resolved pack overheating by correlating SMBus logs with thermal imaging, identifying MOSFET conduction losses and limited shared-drain copper area. Selected lower-RDS(on) devices, enabling the pack to pass subsequent thermal testing.
- Validate PWM, protection thresholds, cutoff timing, and pack performance using oscilloscopes and load testing. Analyze fault/lifetime logs and bench measurements, and present root cause findings to customers.
- Root-caused intermittent high DET-to-GND resistance in a fielded pack through microscope inspection, identifying flex-circuit tears and holes at the Nicomatic Crimpflex connection.
- Develop semi-pack and full-pack assembly and inspection specifications; coordinate with mechanical engineers, manufacturing, PCB suppliers, and customers, including on-site production support in Mexico.
- Design battery electronics to UL 2054 and IEC 62133-2 requirements and support UN 38.3 transport testing. Sustain legacy designs, including documentation and fabrication for medical-pack requalification following a MOSFET end-of-life change.

CERTIFICATION

IPC Certified Interconnect Designer (CID) February 2026

SELECTED PROJECTS

Automated Beverage Dispenser ESP32 | Custom PCB | 3D Printing

- Built an ESP32-controlled dispenser from development-board proof of concept to an integrated, hand-assembled PCB. Combined voltage regulation from a 12 V input, stepper-driven lead-screw motion, and MOSFET-switched peristaltic pumps.
- Modeled and iterated 3D-printed brackets for an aluminum-extrusion frame, refining mechanical rigidity and fit alongside electrical integration.

Rolling Robot Controller KiCad | ESP32 | Motor Control

- Designed a controller PCB for a rolling-robot concept with an ESP32, TB6612FNG motor driver, and MPU6050 IMU, including motor power, sensor interfaces, and decoupling.

TECHNICAL SKILLS

Design: Zuken, KiCad, Altium (familiarity), SPICE, schematic capture, component selection and derating
Lab: Oscilloscopes, logic analyzers, DMMs, milliohmmeters, thermal imaging, electronic loads, soldering
Embedded: ESP32, C/C++, Python, SMBus, I2C, MQTT, TI bqStudio, bq40z50-R2, bq41z90

EDUCATION

Texas A&M University December 2024
B.S. Electrical Engineering | Minor in Mathematics College Station, TX